

ESTHER 1, AB, Canada

WMO: 712400

Lat: 51.670N

Lon: 110.206W

Elev: 707

StdP: 93.11

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.6	-28.6	-37.4	0.1	-30.0	-33.4	0.2	-28.0	13.2	-3.6	11.9	-2.6	2.8	240	0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	31.4	17.5	29.2	16.9	27.2	16.3	19.7	27.3	18.4	25.8	17.3	24.7	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.4	22.8	15.8	12.3	21.0	14.7	11.4	19.8	59.6	27.3	55.0	25.9	51.4	24.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	10.2	9.1	-35.6	34.8	3.3	1.9	-38.0	36.1	-39.9	37.3	-41.8	38.3	-44.2	39.7
			-35.5	21.5	3.2	1.8	-37.8	22.8	-39.7	23.8	-41.5	24.8	-43.9	26.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-12.3	-10.6	-4.1	4.4	10.8	15.0	18.4	17.4	12.2	4.6	-3.9	-10.6	(6)
(7)		DBStd	12.61	9.06	7.95	8.08	5.50	4.51	3.39	3.19	3.55	4.72	5.27	7.22	8.41	(7)
(8)		HDD10.0	3198	693	578	436	179	45	3	0	1	30	178	418	638	(8)
(9)		HDD18.3	5496	951	811	694	418	236	109	37	61	190	425	667	896	(9)
(10)		CDD10.0	834	0	0	1	11	70	153	262	231	96	11	1	0	(10)
(11)		CDD18.3	89	0	0	0	0	2	9	41	32	6	0	0	0	(11)
(12)		CDH23.3	1642	0	0	1	5	86	175	635	558	177	6	0	0	(12)
(13)		CDH26.7	539	0	0	0	0	16	41	225	201	57	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.1	4.2	4.6	4.9	4.9	4.4	4.2	3.9	4.2	4.5	4.2	4.2	(14)
(15)	Precipitation	PrecAvg	330	14	8	13	21	37	73	53	43	25	14	13	11	(15)
(16)		PrecMax	471	47	18	36	61	80	150	128	84	83	33	44	18	(16)
(17)		PrecMin	172	2	1	3	3	3	17	20	5	1	4	0	2	(17)
(18)		PrecStd	75	10	5	7	15	23	32	24	21	21	7	11	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.9	15.8	23.8	28.7	30.5	34.3	34.0	31.9	23.7	14.8	8.3	(19)
(20)			MCWB	3.3	3.4	7.6	11.5	14.1	17.5	19.9	17.7	15.8	12.4	7.5	3.7	(20)
(21)		2%	DB	4.5	4.1	11.8	20.3	25.8	27.5	31.5	31.2	28.0	19.2	10.6	5.5	(21)
(22)			MCWB	1.8	1.5	5.5	10.0	13.2	16.1	18.7	16.9	14.6	10.3	5.6	2.3	(22)
(23)		5%	DB	2.7	2.3	8.7	17.0	23.0	25.3	29.3	28.9	24.9	16.1	7.5	3.4	(23)
(24)			MCWB	0.6	0.2	4.1	8.0	12.3	15.5	17.9	16.7	13.8	8.9	3.8	1.0	(24)
(25)		10%	DB	0.5	0.3	5.9	14.2	20.5	23.1	26.9	26.6	21.9	13.2	5.0	1.0	(25)
(26)			MCWB	-1.1	-1.3	2.6	6.8	11.0	14.5	17.2	16.2	12.5	7.3	2.2	-0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.5	8.5	12.5	16.2	19.9	22.6	20.4	17.0	13.2	8.2	4.6	(27)
(28)			MCDWB	6.3	6.5	14.5	21.8	23.7	27.2	30.2	28.9	27.8	22.7	14.1	7.4	(28)
(29)		2%	WB	2.0	1.7	6.0	10.5	14.5	17.9	20.7	18.8	15.5	10.9	6.1	2.6	(29)
(30)			MCDWB	4.1	3.8	10.8	18.7	23.0	24.3	28.3	27.0	25.7	17.9	10.1	5.0	(30)
(31)		5%	WB	0.6	0.3	4.3	8.8	13.1	16.4	19.1	17.7	14.3	9.2	4.1	1.1	(31)
(32)			MCDWB	2.6	2.0	8.1	15.9	21.5	22.8	26.1	25.7	23.5	15.2	7.3	3.5	(32)
(33)		10%	WB	-1.2	-1.3	2.9	7.2	11.8	15.3	17.9	16.7	13.0	7.7	2.5	-0.8	(33)
(34)			MCDWB	0.5	0.3	5.8	13.1	19.2	21.4	25.1	24.6	21.1	12.6	4.8	1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.5	10.9	13.1	13.8	12.5	14.2	14.6	14.2	12.3	9.8	9.8	(35)
(36)			MCDBR	10.5	11.5	14.4	17.7	18.0	16.2	18.2	19.4	20.1	17.4	12.7	11.2	(36)
(37)			MCWBR	8.6	9.4	8.9	9.1	8.4	7.3	7.4	7.7	9.0	9.6	8.5	8.6	(37)
(38)		5% WB	MCDBR	10.5	11.2	13.5	16.4	16.1	14.3	16.1	16.6	18.5	16.4	12.3	11.2	(38)
(39)			MCWBR	8.8	9.3	8.9	9.0	7.9	7.2	7.5	7.5	8.7	9.4	8.5	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.295	0.300	0.351	0.379	0.373	0.371	0.386	0.331	0.291	0.263	0.251	(40)	
(41)		taud	2.340	2.279	2.414	2.294	2.255	2.333	2.370	2.293	2.427	2.515	2.463	2.353	(41)	
(42)		Ebn at Noon	770	825	905	886	873	880	876	840	856	837	770	719	(42)	
(43)		Edh at Noon	60	86	92	117	128	120	114	116	91	68	53	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.16	2.22	3.60	4.83	5.85	6.04	6.48	5.26	3.78	2.28	1.19	0.86	(44)	
(45)		RadStd	0.12	0.19	0.28	0.41	0.40	0.40	0.29	0.40	0.35	0.29	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page